

APPLE QUALITY PREDICTION FROM CERTAIN FEATURES

Step 1

Data acquisition and processing

Input data

- Training dataset
- Testing dataset

Step 2

Model building

Data preprocessing step

- Data rebalancing
- Harmonization
- Data cleansing
- Normalization
- Scaling

Step 3

Internal and external validation

Step 4

Training step(Algorithm selection)

- Training dataset
- Testing dataset

Step 5

ML Types & Algorithms

Supervised Learning

- Random Forest
- Gradient Boosting (XGBoost)
- SVM/SVC
- K- Nearest Neighbors (KNN)

Step 6

Model Evaluation and Performances check

-Confusion Matrix
-Precision
-Recall
-Accuracy
-AUC
-F1-Score

Step 7

Avoiding Overfitting and Underfitting

-Data Augmentation Process applied
-Cross Validation Technique Applied
-Adding more Features to Datasets
-Done A/B testing

Step 8

Deployment and application

-Exporting the model(Pickle, Joblib)
-Build an API (Flask, Fast, Gradio, Streamlit)
-Deployed on Cloud Server(Docker, AWS)

Prediction Result

Predicted Quality: Good

IDE Used 👍

-VS code
-Jupyter Notebook(Anaconda)

-Google Colab

Core ML/AI Frameworks 👍

-Scikit-learn
-Pandas
-NumPy
-Matplotlib